

Solvency Field Theory V12.7

Binary No-Overwrite, 3+1 Slot Capacity,
and the Mechanical Filter Theorem Stack

From the founding postulate to a nine-mode spectrum
through derived constraints

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June 2026

A stable candidate is a joint-domain-supported winding that can answer the binary no-overwrite question within the 3+1 slot budget.

Publication boundary. This document freezes the V12.7 theorem-stack result. It claims operator-level decomposition of the mechanical seed filter, not a confirmed physical particle spectrum. The field-level derivation of the primitive anchor set, the full PDE/action implementation, Standard Model masses, mass hierarchy, and absolute cadence scale Ω_0 remain open.

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1 The physical picture

What determines which particles can exist?

This question has haunted physics since the discovery that nature uses a specific, seemingly arbitrary set of elementary particles. The Standard Model lists them — six quarks, six leptons, four force carriers, one Higgs boson — but offers no explanation of why *these* particles and not others. The mass spectrum is measured to extraordinary precision and derived from nothing.

V12.6 showed that closure conditions on the normal-bundle holonomy of a maintained worldline produce a *discrete* set of allowed cadence modes rather than a continuum. But V12.6 also showed that pure closure never stops — the rational ladder grows forever. Discreteness without finiteness.

V12.7 provides the finiteness. The mechanism is not a cost cutoff or an energy threshold. It is two derived constraints, both traceable to the founding postulate, that together select exactly nine modes from the infinite closure ladder.

1.1 The jump rope that must pay its own rent

A particle is energy circulating at c in a closed helical path. The winding pattern of the transverse frame — how the internal orientation rotates as the worldline advances — determines the particle's identity. Different winding patterns are different species.

But not every winding pattern can sustain itself. Each tick, the particle writes one Planck area to the boundary. That write must be *distinguishable* from the previous write — otherwise it would overwrite history, violating the founding postulate. And the winding pattern must *fit* in the available transverse structure of the worldline — the geometry of 3+1 dimensional spacetime limits what patterns are physically realizable.

These two requirements — distinguishability and dimensional fit — are the two branches of the mechanical filter.

1.2 Why the half-turn

Each write to the boundary has an angular orientation — the transverse frame direction at the moment of inscription. The no-overwrite rule requires that consecutive writes be distinguishable. But distinguishability on a compact circle requires a minimum angular separation.

No-overwrite asks one question: “is this write the same as the previous write, or different?” That question is binary. The answer is yes or no. A binary question requires exactly two distinguishable states. On a compact circle, two equal states occupy half the circle each — a diametric partition, with each write claiming one semicircle as its footprint.

Two semicircular footprints are distinguishable if and only if they don't overlap. Non-overlap on a unit circle requires angular separation of at least $1/2$. Since the cadence ratio ρ is the angular advance per base cycle, the distinguishability condition is:

$$\boxed{\rho \geq \frac{1}{2}} \tag{1}$$

This threshold is not inserted. It is derived from three facts: no-overwrite is binary, binary on a circle is diametric, and diametric footprints require half-turn separation.

Two independent arguments support the half-turn. The first (R92A) is geometric: the diametric partition is the unique antipodal binary partition with no gap and no overlap. The second (R92B) is information-theoretic: $K = 2$ is the minimum nontrivial compact-phase record, giving cell width $1/K = 1/2$. These two arguments were shown to be the same theorem in two languages (R93): the angular Planck bit *is* the antipodal binary partition.

Every control was defeated. $K = 1$ cannot encode same/different. $K > 2$ over-refines the predicate and admits intruders. Point writes accept everything. Modulo-only checks erase winding history. Exact-antipode-only checks are too strict. The half-turn is the unique threshold that is neither too loose nor too strict.

1.3 Why four slots

A winding pattern with denominator N — meaning the transverse frame takes N base cycles to close — requires N independently maintainable phase slots over one closure period. The worldline tube in 3+1 dimensions supplies a limited number of independent channels: one tangent/base cadence channel (the Compton cycle itself) and three transverse normal-frame channels (one for each spatial dimension perpendicular to the worldline).

Total slot capacity:

$$\boxed{N_{\text{slots}} = 1 + 3 = 4.} \quad (2)$$

A mode with $N > 4$ requires more independently maintainable slots than the spacetime geometry can provide. The constraint is:

$$\boxed{N \leq 4.} \quad (3)$$

This is not an energy cutoff. It is a dimensional capacity limit — 3+1 spacetime literally does not have room for winding patterns that need more than four independent phase channels.

Again, every control was defeated. Capacity 3 loses valid modes. Three normal channels without the tangent channel also lose modes (proving the base cadence channel counts). Capacity 5 admits intruders ($3/5$, $4/5$). Unbounded capacity produces runaway growth. A bare $N \leq 4$ without the 3+1 decomposition reproduces the numbers but lacks the native origin.

1.4 Why these nine

The third branch is the support domain. Not every mode with $\rho \geq 1/2$ and $N \leq 4$ participates in the spectrum — only those that are connected to the closure network through reflection ($\rho \mapsto 2 - \rho$) and reciprocal ($\rho \mapsto 1/\rho$) maps.

The minimal joint support domain is generated from three primitive anchors — $\{1, 1/2, 1/3\}$ — under the reflection and reciprocal maps. The anchor 1 is the identity cadence. The anchor $1/2$ is the binary no-overwrite cell. The anchor $1/3$ is the tri-normal virtual support (the minimal non-self-conjugate residue from R18's $N = 3$ selection), which is not itself a stable species but generates closure partners that are.

The combined operator selects the nine modes that satisfy all three conditions simultaneously:

$$\boxed{S_{12.7} = \{\rho \in D_{RC} : \rho \geq \frac{1}{2}, \text{denom}(\rho) \leq 4\} = \{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, 1, \frac{5}{4}, \frac{4}{3}, \frac{3}{2}, \frac{5}{3}, 2\}.} \quad (4)$$

Nine modes. Two derived constraints. One support domain. Zero free parameters. The seed remains exact while the unconstrained closure ladder grows to 50+ modes. Every weakening of the filter admits intruders. Every strengthening kills valid modes.

1.5 What the seed is and is not

The nine-mode seed is a *structural candidate roster*, not a confirmed particle spectrum. Mapping these cadence ratios to real measured masses requires the absolute cadence scale Ω_0 , envelope normalization, and a physical identification of which modes correspond to which particles. None of these are derived.

What V12.7 establishes is the *mechanism for finiteness*: why a specific, small, finite number of modes survive from an infinite ladder. The finiteness is not imposed by a cutoff. It is forced by the intersection of no-overwrite distinguishability, dimensional capacity, and closure support. The particles that exist — if this scaffold is correct — are the ones that can answer a yes/no question on a compact phase circle in 3+1 dimensions.

2 The half-turn branch: binary no-overwrite

2.1 The proof chain

The derivation proceeds in six steps, each tested by an independent gate:

1. **No-overwrite is binary** (R94A/B). No-overwrite asks “same or different?” — a predicate with exactly two truth values.
2. **Minimum faithful record is $K = 2$** (R94A/B). A binary predicate requires at least two distinguishable states. $K = 1$ cannot distinguish. $K > 2$ answers an unasked question.
3. **Half-turn cell width** (R92A/B, R93). $K = 2$ equal cells on a compact circle each occupy width $1/K = 1/2$. This is simultaneously the unique antipodal binary partition (no gap, no overlap) and the minimum nontrivial angular record (the angular Planck bit). The two arguments fuse into one theorem.
4. **Non-overlap update** (R95). A new write is accepted if and only if its unwrapped binary cell does not overlap the previous write cell: $D(n+1, n) = 1 \iff I_{n+1} \cap I_n = \emptyset$.
5. **Threshold recovery** (R90–R91, R95). For $K = 2$, cell width $1/2$, the non-overlap condition gives $\rho \geq 1/2$. The threshold is derived, not inserted.
6. **Control verification** (R90, R95). Every alternative was tested and defeated. The half-turn is the unique surviving threshold.

2.2 The micro-update operator

The operator is:

$$D(n+1, n) = 1 \iff I_{n+1} \cap I_n = \emptyset, \quad (5)$$

where $I_n = [\theta_n, \theta_n + 1/2)$ is the binary write cell on the unwrapped phase history. This is the non-overlapping semicircle check — the simplest possible implementation of a binary distinguishability test on a compact circle.

3 The slot-capacity branch: 3+1 dimensional capacity

A denominator- N mode requires N independently maintainable write slots per closure cycle. A timelike worldline tube in 3+1 spacetime supplies:

- 1 tangent/base cadence channel (the Compton cycle),
- 3 transverse normal-frame channels (one per perpendicular spatial dimension).

Total: $1 + 3 = 4$. Therefore $N \leq 4$.

The base cadence channel is required: three normal channels alone (capacity 3) lose $N = 4$ seed modes. An extra dimension (capacity 5) admits $N = 5$ intruders. The fourth slot is specifically the tangent/base cadence channel, not an arbitrary extra.

4 The closure branch: joint support domain

The support domain is generated from three primitive anchors under the reflection map $R(\rho) = 2 - \rho$ and the reciprocal map $C(\rho) = 1/\rho$:

$$D_{RC} = \text{closure}_{R,C}(\{1, \frac{1}{2}, \frac{1}{3}\}). \quad (6)$$

The three anchors have native interpretations:

- 1 = identity cadence anchor (the base winding),
- $1/2$ = binary no-overwrite cell anchor (from the half-turn derivation),
- $1/3$ = tri-normal virtual support anchor (minimal non-self-conjugate residue; not itself a stable species).

R98–R99 established that the correct formulation is *joint support-domain membership*, not an irreducible per-mode dual-AND condition. Both reflection and reciprocal maps are needed to *generate* the domain, but once generated and filtered by $\rho \geq 1/2$ and $N \leq 4$, domain membership, R -partner support, C -partner support, and dual-partner support all select the same nine modes.

Controls: R -only generation gives only 4 modes (too few). C -only generation gives only 3 modes. Omitting the $1/3$ anchor loses $5/3$. Omitting the identity anchor loses 1. Adding $1/4$ as an anchor admits $7/4$ as an intruder. The anchor set $\{1, 1/2, 1/3\}$ is the minimal set that generates exactly the nine-mode seed.

5 The combined operator and the nine-mode seed

The frozen V12.7 operator is:

$$\boxed{\text{accept} \iff \rho \in D_{RC}, \quad \rho \geq \frac{1}{2}, \quad \text{denom}(\rho) \leq 4.} \quad (7)$$

The recovered seed is:

$$\boxed{\frac{1}{2}, \quad \frac{2}{3}, \quad \frac{3}{4}, \quad 1, \quad \frac{5}{4}, \quad \frac{4}{3}, \quad \frac{3}{2}, \quad \frac{5}{3}, \quad 2.} \quad (8)$$

Interval stress tests across denominator scans from 6 to 32 confirm: the nine-mode seed remains exact while unconstrained closure grows to 50+ modes.

Every weakening of the filter admits intruders or causes growth. Every strengthening kills valid modes. The exact filter is sharp.

Control	Modes	Targets	Result
Point threshold, cap 4	10	5/5	Admits 1/3 intruder
Half threshold, cap 3	7	4/5	Loses $N=4$ modes
Half threshold, cap 4	9	5/5	Exact seed
Half threshold, cap 5	13	5/5	$N=5$ intruders
Half threshold, unbounded	49	5/5	Runaway growth
2/3 threshold, cap 4	8	4/5	Loses 1/2 mode
R -only domain	4	3/5	Domain too small
C -only domain	3	2/5	Domain too small
Omit 1/3 anchor	8	4/5	Loses 5/3
Omit identity anchor	8	4/5	Loses 1
Add 1/4 anchor	10	5/5	Admits 7/4

Layer	Name	Statement	Status
1	No-overwrite predicate	No-overwrite asks same/different, a binary predicate.	Pass
2	Minimum faithful record	Minimum faithful compact-phase record alphabet is $K=2$.	Pass
3	Half-turn cell	$K=2$ gives cell width $1/K = 1/2$.	Pass
4	Binary no-overlap update	New write accepted iff unwrapped binary cell does not overlap prior.	Pass
5	Half-turn criterion	For $K=2$, no-overlap gives $\rho \geq 1/2$.	Pass
6	3+1 slot capacity	One base/tangent + three normal channels = capacity 4.	Pass
7	Denominator criterion	Denominator- N mode needs N slots, so $N \leq 4$.	Pass
8	Joint R/C support domain	Stable candidates lie in $\text{closure}_{R,C}(\{1, 1/2, 1/3\})$.	Candidate pass
9	Mechanical seed	Joint-domain + $\rho \geq 1/2$ + $N \leq 4 \rightarrow$ exact 9-mode seed.	Stack pass

6 The theorem stack

7 What is earned, conditional, and open

7.1 Earned

The $\rho \geq 1/2$ rule traces to binary no-overwrite distinguishability.

The $N \leq 4$ rule traces to 3+1 dimensional slot capacity.

The closure branch traces to the minimal joint R/C support domain from $\{1, 1/2, 1/3\}$.

The combined operator recovers the exact 9-mode seed under interval stress while every control fails in the predicted direction.

The seed filter is no longer a post-hoc numerical cut. It is decomposed into theorem-sized operator layers.

7.2 Conditional

The three primitive anchors $\{1, 1/2, 1/3\}$ have native interpretations (identity, binary cell, tri-normal virtual support) but their field-level derivation remains open.

The joint support domain is operator-level, not field-level.

7.3 Open

Field-level derivation of the primitive anchor set.

Full PDE/action implementation of the update and domain operators.

Mapping the 9 toy cadence modes to real particle species.

Standard Model mass hierarchy.

Absolute cadence scale Ω_0 .

Envelope normalization and coupling constants.

7.4 Rejected routes

Post-hoc numerical cut: rejected. The filter is now decomposed into derived operator layers.

Proxy self-solvency (R87): rejected. Feature-blending proxies do not capture the physics; mechanical constraints do.

Irreducible per-mode dual-AND support: rejected by R98/R99. The correct statement is joint support-domain membership.

Pure closure without truncation: rejected by R86. Closure alone grows forever. Finiteness requires the derived constraints.

8 Conclusion

V12.7 answers the question: why is the closure spectrum finite?

The answer is two constraints, both derived:

$$\rho \geq \frac{1}{2} \quad \text{from binary no-overwrite,} \quad N \leq 4 \quad \text{from 3+1 dimensional capacity,} \quad (9)$$

combined with membership in the minimal joint R/C support domain.

The nine surviving modes are not chosen by fitting. They are the only modes that satisfy all three conditions simultaneously. Every weakening of the filter admits intruders. Every strengthening kills valid modes. The filter is sharp.

Whether these nine modes correspond to real particles remains the central open question. The absolute cadence scale Ω_0 , the mass mapping, and the Standard Model identification are the frontier beyond V12.7. But the mechanism for finiteness — the reason *some* discrete set exists at all, and the constraints that determine its size — is now decomposed into theorem-sized pieces traceable to the founding postulate.

The particles that exist are the ones that can answer a yes/no question on a compact phase circle in 3+1 dimensions.

Acknowledgments

The R88–R99 FORCE audit sequence was developed and executed in collaboration with ChatGPT/5.5 (OpenAI). The physical narrative and the binary-predicate argument (R94B, R92B) were developed with Claude/Opus (Anthropic). The theoretical framework is SFT [1, 2, 3, 4, 5, 6].

References

- [1] S. Fosmark, *Solvency Field Theory V11: Master Framework*, Zenodo (2026). <https://zenodo.org/records/20350797>
- [2] S. Fosmark, *SFT V12.1: The One-Shatter Closure*, Zenodo (2026).
- [3] S. Fosmark, *SFT V12.3: Gravity Trajectory Prevalidation*, Zenodo (2026).
- [4] S. Fosmark, *SFT V12.4: Force-Sector and Electroweak Bridge*, Zenodo (2026).
- [5] S. Fosmark, *SFT V12.5: The Quantum Bridge*, Zenodo (2026).
- [6] S. Fosmark, *SFT V12.6: Closure Spectrum Scaffold*, Zenodo (2026).

A Gate rollup: R88–R99

Gate	Title	Contribution	Status
R88	Mechanical filter	Recovered 9-mode seed with dual support + $\rho \geq 1/2$ + $N \leq 4$.	Pass
R89	Native rule candidates	3+1 slot-budget and half-turn origins survived first audit.	Pass
R90	Half-turn uniqueness	$\rho \geq 1/2$ is the unique threshold: weaker admits intruders, stronger kills.	Pass
R91	Footprint conditional	Given half-turn footprint, no-overwrite $\iff \rho \geq 1/2$.	Pass
R92A	Antipodal partition	Half-turn is the unique binary partition with no gap, no overlap.	Pass
R92B	Angular Planck bit	$K=2$ minimum nontrivial record gives width $1/2$.	Pass
R93	Fusion	R92A and R92B are one theorem in two languages.	Pass
R94A	Predicate quotient	$K > 2$ is harmful overstructure or redundant quotient.	Pass
R94B	Binary predicate	No-overwrite is yes/no; $K=2$ is minimum faithful record.	Pass
R95	Micro-update operator	Binary unwrapped no-overlap update derives $\rho \geq 1/2$. Every control defeated.	Pass
R96	Slot capacity	$1+3 = 4$ channels from 3+1 worldline tube. Every control defeated.	Pass
R97	Theorem-stack rollup	Combined filter recovers exact seed; all controls fail as predicted.	Pass
R98	Support-domain origin	Joint R/C closure from $\{1, 1/2, 1/3\}$ recovers exact seed.	Candidate
R99	Language patch	Joint support-domain membership, not per-mode dual-AND.	Pass